FtsZ-Independent Cell Division Mechanisms in Bacteria and Archaea: Evolutionary Anomalies and Primitive Division Pathways

A Comprehensive Review of Alternative Cellular Division Strategies

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Abstract

The tubulin homolog FtsZ is universally conserved across most bacterial lineages and is considered essential for binary fission. However, several prokaryotic lineages have evolved FtsZ-independent cell division mechanisms that challenge this paradigm. This research note summarizes current knowledge of these alternative division pathways, including: (1) Chlamydiae using MreB-dependent polarized budding, (2) Crenarchaeota employing ESCRT-III-mediated division, and (3) L-form bacteria utilizing membrane-based proliferation. These mechanisms represent not only evolutionary exceptions but potential windows into primitive cellular division strategies that may have operated early in the evolution of life. The existence of multiple, independently evolved FtsZ-independent pathways suggests that the Last Universal Common Ancestor (LUCA) may have possessed more flexible division capabilities than previously assumed.

Keywords: *FtsZ-independent division, Chlamydia, archaeal division, ESCRT-III, L-forms, cell evolution*

1. Introduction

Cell division is a fundamental process for all cellular life. In bacteria, this process is typically orchestrated by the divisome, a protein complex centered on the tubulin homolog FtsZ. FtsZ polymerizes into a dynamic Z-ring at the future division site, recruits downstream proteins, and directs peptidoglycan synthesis to achieve binary fission. This mechanism is so highly conserved that FtsZ was considered indispensable for bacterial cytokinesis.

However, discoveries over the past two decades have revealed several prokaryotic lineages that successfully divide without FtsZ. These findings challenge our understanding of cell division evolution and suggest that alternative mechanisms may be more widespread than previously appreciated. The three best-characterized FtsZ-independent systems are: (1) Chlamydiae: MreB-dependent polarized budding, (2) Crenarchaeota: ESCRT-III-mediated membrane scission, and (3) L-form bacteria: Membrane blebbing and tubulation.

This review synthesizes current knowledge of these alternative mechanisms and discusses their evolutionary significance.

2. Chlamydiae: MreB-Dependent Polarized Budding

Members of the phylum Chlamydiae represent one of the most striking exceptions to FtsZ-dependent division. These obligate intracellular pathogens have completely eliminated the ftsZ gene from their reduced genomes. Instead, they employ a unique MreB-dependent polarized division process that resembles budding rather than binary fission.

2.1 The MreB-RodZ Complex

Chlamydial MreB, an actin homolog, forms a complex with RodZ to create a functional replacement for the FtsZ-based divisome. This complex directs both cell division and peptidoglycan synthesis in a spatially coordinated manner. Unlike the symmetric Z-ring formation in typical bacteria, Chlamydia exhibits unipolar growth with division occurring at one pole of the cell.

2.2 Polarized Division Process

The division process in Chlamydia trachomatis proceeds through several distinct stages. First, FtsK, a DNA translocase protein, initiates assembly of a unique divisome complex at the cell pole. Second, unipolar growth extends the cell body, creating asymmetric expansion. Third, cardiolipin concentrates at the poles and septum, promoting MreB recruitment. Finally, the daughter cell separates through a mechanism independent of FtsZ constriction. This polarized budding mechanism is fundamentally different from the symmetric binary fission used by most bacteria and more closely resembles budding processes observed in Planctomycetes.

3. Crenarchaeota: ESCRT-III-Mediated Division

The entire phylum Crenarchaeota lacks FtsZ and instead employs the ESCRT-III (Endosomal Sorting Complexes Required for Transport III) system for cell division. This finding was particularly surprising because ESCRT-III was previously known only from eukaryotes, where it functions in membrane scission during processes like cytokinesis, viral budding, and multivesicular body formation.

3.1 The CdvAB Complex

In Sulfolobus species, the primary division machinery consists of three components: CdvA, a conserved protein that recruits ESCRT-III to membranes via a peptide-winged helix domain; CdvB, the ESCRT-III homolog that forms the core fission machinery; and CdvC, a Vps4 ATPase homolog that disassembles the ESCRT-III filaments. These proteins first form a non-contractile ring at the midcell, establishing the division plane before membrane scission.

3.2 Membrane Scission Mechanism

The ESCRT-III system assembles a membrane neck structure during division, functioning as the minimal membrane fission machinery. Recent studies have revealed a "relay race" mechanism where multiple ESCRT-III paralogs sequentially drive the division process, with proteasomal degradation providing temporal control. The archaeal ESCRT-III system represents a simplified version of the eukaryotic machinery, suggesting that this division mechanism may have been present in the last common ancestor of archaea and eukaryotes.

4. L-Form Bacteria: Membrane-Based Proliferation

L-forms are cell wall-deficient bacterial variants that can proliferate without the peptidoglycan synthesis machinery typically required for division. When cell wall synthesis is inhibited (either genetically or pharmacologically), some bacteria can switch to an alternative proliferation mode based entirely on membrane dynamics.

4.1 Membrane Blebbing and Tubulation

L-form division occurs through several membrane-based processes: blebbing (formation and expansion of membrane protrusions), tubulation (extension of membrane tubes), vesiculation (pinching off of membrane vesicles), and fission (separation of daughter cells). These processes are completely independent of both FtsZ and peptidoglycan synthesis.

4.2 Biophysical Requirements

Successful L-form proliferation depends on: (1) Increased surface area-to-volume ratio, which promotes membrane curvature changes; (2) Osmoprotective conditions, required to compensate for cell wall absence; (3) Metabolic adaptation, as cell wall synthesis blockade creates metabolic imbalances that must be compensated; and (4) Oxidative stress management, with cell growth limited by oxidative stress responses. The L-form mechanism may represent a primitive form of cell division that could have operated before the evolution of cell walls, providing insights into early cellular life.

5. Evolutionary Implications

The existence of multiple, independently evolved FtsZ-independent division mechanisms has profound implications for our understanding of cell evolution.

5.1 LUCA's Division Machinery

The presence of FtsZ in most bacteria and many archaea suggests it was present in the Last Universal Common Ancestor (LUCA). However, the existence of multiple alternative mechanisms suggests LUCA may have possessed more flexible division capabilities, with different lineages subsequently specializing in different strategies.

5.2 Evolutionary Plasticity

The independent evolution of FtsZ-independent mechanisms in Chlamydiae (bacteria), Crenarchaeota (archaea), and L-forms (facultative state) demonstrates remarkable evolutionary plasticity in cell division mechanisms. This suggests that the constraint against evolving alternative division strategies is weaker than previously assumed.

5.3 Primitive Division Pathways

The L-form membrane-based proliferation mechanism, in particular, may represent a primitive division strategy that could have operated in early cells before the evolution of complex cell walls or cytoskeletal elements. This has implications for understanding the earliest stages of cellular evolution.

6. Future Directions

Several key questions remain unanswered: (1) How widespread are FtsZ-independent mechanisms? Metagenomic surveys may reveal additional lineages that have lost FtsZ. (2) What are the molecular details of Chlamydial division? The complete composition of the MreB-based divisome requires further characterization. (3) How did ESCRT-III evolve? The evolutionary origins of the archaeal ESCRT-III system and its relationship to eukaryotic systems need clarification. (4) Can FtsZ-independent mechanisms be targeted therapeutically? These alternative pathways could represent novel antibiotic targets.

7. Conclusions

FtsZ-independent cell division mechanisms are no longer mere curiosities but represent fundamental alternative strategies for cellular propagation. The three systems described here—MreB-dependent polarized budding in Chlamydiae, ESCRT-III-mediated division in Crenarchaea, and membrane-based proliferation in L-forms—demonstrate the evolutionary flexibility of cell division mechanisms.

These findings challenge the notion that FtsZ is indispensable for bacterial cytokinesis and suggest that our understanding of cell division evolution remains incomplete. Future research integrating comparative genomics, structural biology, and biophysics will be essential for elucidating the full diversity of cellular division strategies and their evolutionary significance.

References

[1] Bi E, Lutkenhaus J. (1991). FtsZ ring structure associated with division in Escherichia coli. Nature, 354: 161-164.

[2] Goley ED, Wickner S. (2021). FtsZ dynamics in bacterial division: what, how, and why? J Mol Biol, 433: 166736.

[3] Bernardet, et al. (2020). Division without Binary Fission: Cell Division in the FtsZ-Less... J Bacteriol, 2020.

[4] Jacquier N, et al. (2015). The MreB-RodZ complex in Chlamydia. PMC7029139.

[5] Kumar M, et al. (2021). Localized peptidoglycan biosynthesis in Chlamydia. Front Microbiol, 12: 658732.

[6] Pilhofer M, et al. (2020). Polarized cell division in Chlamydia. PLOS Pathog.

[7] Bernander R, et al. (2000). Cell division in archaea: phylogenetic diversity and similarities. Mol Microbiol, 37: 4.

[8] Lindas AC, et al. (2008). The Cdv complex in archaeal cell division. PMC2686351.

[9] Samson RY, et al. (2011). ESCRT-III mediated cell division in archaea. Nature, 474: 4.

[10] Tarrago R, et al. (2020). ESCRT-III paralogs drive cell division. Mol Cell, 78: 9.

[11] Dominguez-Cuevas P, et al. (2012). Formation and proliferation of wall-free bacteria. eLife, 3: 04629.

[12] Errington J. (2013). L-form bacteria, cell walls and the origins of life. PMC3603455.

[13] Mercier R, et al. (2020). Metabolic limitations on L-form cell growth. Curr Biol, 25: 1331-1336.

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